**Contact**
Department of Biology
Tufts University
200 College St
Medford, MA 02155
`Lawrence.Uricchio@tufts.edu`
`https://scholar.google.com/citations?user=XyVUpZYAAAAJ`
`uricchio.github.io`

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| **Positions** | 2021- | *Assistant Professor*, Department of Biology, Tufts University |
| **Education** | 2018-2021 | *Postdoctoral Scholar*, Integrative Biology, University of California, Berkeley<br>Advisors: Dr. Mike Boots & Dr. Becca Tarvin |
| | 2015-2018 | *Postdoctoral Fellow*, Biology, Stanford University<br>Advisors: Dr. Noah Rosenberg & Dr. Erin Mordecai |
| | 2014 | *PhD*, Bioinformatics, University of California, San Francisco<br>PhD advisor: Dr. Ryan Hernandez |
| | 2011 | *MS*, Computer Science, University of Chicago |
| | 2009 | *MS*, Biophysical Sciences, University of Chicago |
| | 2005 | *BA*, Physics, Carleton College |
| **Honors** | 2021- | Youniss Family Assistant Professor of Innovation |
| | 2016-2018 | NIH IRACDA Postdoctoral Fellow, Stanford University & SJSU |
| | 2015-2016 | CEHG Postdoctoral Fellow, Stanford University |
| | 2014 | Discovery Fellow, UCSF |
| | 2014 | ASHG trainee award for excellence in human genetics research, semi-finalist |
| | 2013 | Teaching Assistant Excellence Award, UCSF |
| | 2012-2014 | Achievement Rewards for College Scientists Fellow, UCSF |
| | 2005 | Distinction in the physics major, Carleton College |
| | 2005 | Phi Beta Kappa (academic honor society), Carleton College |
| | 2005 | Sigma Xi (scientific research honor society), Carleton College |
| | 2005 | Magna Cum Laude, Carleton College |
| | 2002 | Dean's list, Carleton College |

## Articles & conference papers († *: corresponding author;* ∗*: equal contributions*)

*In progress*

[1] Wright-Lichter JX, Leav A, Gordon S, **Uricchio LH**† (2025) Multimodal selection and the evolution of plumage on a warming planet. **submitted**.

[2] Dabholkar S, Gouvea J, Coburn K, **Uricchio LH**. Using "Resistance" Spotlights to Support Studentsâ Critical Reflections on and Reimaginations of Disciplinary Values and Participation. **in revision**.

[3] Lele K†, **Uricchio LH**† (2026) Fitness landscapes for species interactions: when do population genetics and adaptive dynamics diverge?. *bioRxiv,* **submitted**.

[4] Calderon A, Kozak GM, Uricchio LH† (2026) Leveraging the joint site frequency spectrum to detect genomic regions of early divergence. *bioRxiv,* **submitted**.

*Peer reviewed*

[1] Lele K, Wolfe BE, **Uricchio LH**† (2026) Pairwise interactions and serial bottlenecks help explain species composition in a multispecies microbial community. *Ecology,* **107**(1), e70284.

[2] Wuitchik DM, Fifer JE, Huzar AK, Pechenick JA, **Uricchio L**, Davies SW (2025) Outside your shell: exploring genetic variation in two congeneric marine snails across a latitude and temperature gradient. *BMC Genomics,* **26**, 486.

[3] Gearty W, **Uricchio LH**, Lyons K (2024) Investigating the Biotic and Abiotic Drivers of Body Size Disparity in Communities of Non-Volant Terrestrial Mammals. *Global Ecology and Biogeography,* p. e13913.

[4] Murga-Moreno J, Casillas S, Barbadilla A, **Uricchio L***†, Enard D*† (2024) An efficient and robust ABC approach to infer the rate and strength of adaptation. *G3: Genes, Genomes, Genetics,* **14**(4), jkae031.

[5] Louw N, Wolfe BE, **Uricchio LH**†. (2024) A phylogenomic perspective on interspecific competition. *Ecology Letters,* **27**(2), e14359.

[6] Dabholkar S, Gouvea J, **Uricchio LH**. (2023) Designing for and Characterizing Critical Navigation of Disciplinary Values in an Undergraduate Computational Biology Course. In *Proceedings of the 17th International Conference of the Learning Sciences-ICLS 2023, pp. 1386-1389* International Society of the Learning Sciences.

[7] Wan YC, Navarrete Méndez M, O'Connell LA, and **Uricchio LH**, and Roland AB, Maan ME Ron SR, Betancourth-Cundar M et al. (2023) Selection on visual opsin genes in diurnal Neotropical frogs and loss of the SWS2 opsin in poison frogs. *Molecular Biology and Evolution,* **40**(10), msad206.

[8] **Uricchio LH**, Bruns EL, Hood M, Boots M, Antonovics J. (2023) Multimodal pathogen transmission as a limiting factor in host distribution. *Ecology,* **104**(3), e3956.

[9] Visher E, **Uricchio L**, Bartlett L, DeNamur N, Yarcan A, Alhassani D, Boots M. (2022) The evolution of host specialization in an insect pathogen. *Evolution,* **76**(10), 2375–2388.

[10] Ebel ER, **Uricchio LH**, Petrov DA, Egan ES. (2022) Revisiting the malaria hypothesis: accounting for polygenicity and pleiotropy. *Trends in Parasitology,* **38**(4).

[11] Couper LI, Farner JE, Caldwell JM, Childs ML, Harris MJ, Kirk DG, Nova N, Shocket M, Skinner EB, **Uricchio LH**, Exposito-Alonso M, Mordecai EA. (2021) How will mosquitoes adapt to climate warming?. *eLife,* **10**, e69630.

[12] Song C, **Uricchio LH**, Mordecai EA, Saavedra S. (2021) Understanding the emergence of contingent and deterministic exclusion in multispecies communities. *Ecology Letters,* **24**, 2155â2168.

[13] **Uricchio LH**†. (2020) Evolutionary perspectives on polygenic selection, missing heritability, and GWAS. *Human Genetics,* **139**(1), 5–21.

[14] Severson AL, **Uricchio LH**, Arbisser IM, Glassberg EC, Rosenberg NA. (2019) Analysis of author gender in TPB, 1991-2018. *Theoretical Population Biology,* **127**, 1–6.

[15] **Uricchio LH**, Petrov DA, Enard D. (2019) Exploiting selection at linked sites to infer the rate and strength of adaptation. *Nature Ecology & Evolution,* **3**(6), 977–984.

[16] Hernandez RD, **Uricchio LH**, Hartman K, Ye C, Dahl A, Zaitlen N. (2019) Ultrarare variants drive substantial cis heritability of human gene expression. *Nature Genetics,* **51**(9), 1349–1355.

[17] **Uricchio LH***, Kitano HC*, Gusev A, Zaitlen NA. (2019) An evolutionary compass for detecting signals of polygenic selection and mutational bias. *Evolution Letters,* **3**(1), 69–79.

[18] **Uricchio LH**†, Daws SC, Spear ER, Mordecai EA†. (2019) Priority effects and nonhierarchical competition shape species composition in a complex grassland community. *The American Naturalist,* **193**(2), 213–226.

[19] Gignoux CR, Torgerson DG, Pino-Yanes M, **Uricchio LH**, Galanter J, Roth LA, Eng C, Hu D, Nguyen EA, Huntsman S et al. (2019) An admixture mapping meta-analysis implicates genetic variation at 18q21 with asthma susceptibility in Latinos. *Journal of Allergy and Clinical Immunology,* **143**(3), 957–969.

[20] Goldberg A, **Uricchio LH**, Rosenberg NA (2018) Natural selection in human populations. *Oxford Bibliographies in Evolutionary Biology,* DOI:10.1093/OBO/9780199941728-0112.

[21] **Uricchio LH**, Warnow T, Rosenberg NA (2016) An analytical upper bound on the number of loci required for all splits of a species tree to appear in a set of gene trees. *BMC Bioinformatics,* **17**(14), 241–250.

[22] **Uricchio LH**†, Zaitlen NA, Ye CJ, Witte JS, Hernandez RD†. (2016) Selection and explosive growth alter genetic architecture and hamper the detection of causal rare variants. *Genome Research,* **26**, 863–873.

[23] **Uricchio LH**, Torres R, Witte JS, Hernandez RD. (2015) Population genetic simulations of complex phenotypes with implications for rare variant association tests. *Genetic Epidemiology,* **39**(1), 35–44.

[24] **Uricchio LH**, Hernandez, RD. (2014) Robust forward simulations of recurrent hitchhiking. *Genetics,* **197**(1), 221–236.

[25] Maher MC*, **Uricchio LH***, Torgerson DG, Hernandez RD. (2013) Population genetics of rare variants and complex diseases. *Human Heredity,* **74**(3-4), 118–128.

[26] **Uricchio LH**, Chong JX, Ross KD, Ober C, Nicolae DL. (2012) Accurate imputation of rare and common variants in a founder population from a small number of sequenced individuals. *Genetic Epidemiology,* **36**(4), 312–319.

[27] Çalışkan M, Chong JX, **Uricchio L**, Anderson R, Chen P *et al.* (2011) Exome sequencing reveals a novel mutation for autosomal recessive nonsyndromic mental retardation in the TECR gene on chromosome 19p13. *Human Molecular Genetics,* **20**(7), 1285–1289.

[28] Miller LS, Pietras EM, **Uricchio LH**, Hirano K, Rao S *et al.* (2007) Inflammasome-mediated production of IL-1$\beta$ is required for neutrophil recruitment against Staphylococcus aureus in vivo. *Journal of Immunology,* **179**(10), 6933–6942.

[29] Pattanayak AK, Brooks DWC, de La Fuente A, **Uricchio L**, Holby E *et al.* (2005) Coarse-grained entropy decrease and phase-space focusing in Hamiltonian dynamics. *Physical Review A,* **72**(1), 013406.

*Dissertation*

[1] **Uricchio LH**. (2014) Models and forward simulations of selection, human demography, and complex traits. UNIVERSITY OF CALIFORNIA, SAN FRANCISCO.

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| **Teaching** | 2026 | Instructor, *Computational Laboratory in Population Genomics*, Bio 145, Tufts University |
| | 2025 | Instructor, *Computational Biology*, Bio 35, Tufts University |
| | 2025 | Instructor, *Computational Laboratory in Population Genomics*, Bio 145, Tufts University |
| | 2024 | Instructor, *Computational Biology*, Bio 35, Tufts University |
| | 2022 | Guest lecture, *R and Programming for STEM*, Bio 196, Tufts University |
| | 2022 | Instructor, *Biostatistics*, Bio 132, Tufts University |
| | 2022 | Instructor, *Computational Biology*, Bio 35, Tufts University |
| | 2022 | Instructor, *Seminar in Ecology and Evolution*, Bio 244, Tufts University |
| | 2022 | Guest lecture, *Resilience seminar*, Bio 263, Tufts University |
| | 2021 | Instructor, *Computational Biology*, Bio 35, Tufts University |
| | 2018 | Co-instructor, *Evolutionary genetics*, San Jose State University |
| | 2017 | Co-instructor, *Ecology*, San Jose State University |
| | 2017 | Co-founder, *Stanford postdoc pedagogy miniseries*, Stanford University |
| | 2016-2017 | Guest lecturer, *Evolution*, Stanford University |
| | 2014 | Graduate teaching assistant, *Computational Evolutionary Genomics*, UCSF |
| | 2013 | Graduate student instructor, *Computational Biology*, UC Berkeley |
| | 2009 | Guest lecturer, *Genes, Networks, and Cells*, University of Chicago |
| | 2003-2004 | Teaching assistant, tutor, & grader, *Introduction to Physics, Classical & Computational Mechanics, Contemporary Experimental Physics*, Carleton College |

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| **Service** | 2026 | NSF Panelist (DEB) |
| | 2025-2026 | Tufts Biology Department Graduate Admissions Committee |
| | 2025-2026 | Tufts University Faculty DEIJ Committee |
| | 2024-2025 | Tufts University Faculty Committee on Work/Life |
| | 2024 | NSF Panelist (DEB) |
| | 2023 | Tufts Biology DEIJ Co-chair, spring semester |
| | 2023 | NSF *ad hoc* grant reviewer |
| | 2022 | Canada Foundation for Innovation Grant Reviewer |
| | 2021 | NSF panelist (IOS) |
| | 2022 | Committee Member, lecturer search committee, Tufts Biology Department |
| | 2021-2025 | Committee Member, Tufts Biology DEIJ committee (on leave for 2023-2024) |
| | 2013- | Invited peer reviewer for *Genetics*, *BMC Evolutionary Biology*, *PLoS Genetics*, *Molecular Biology & Evolution*, *Genome Biology & Evolution*, *Molecular Ecology*, *Heredity*, *Theoretical Population Biology*, *Nature Genetics*, *American Journal of Human Genetics*, *Entropy*, *Functional Ecology*, *Peer Community in Evolutionary Biology*, *Journal of Allergy and Clinical Infection*, *eLife*, *PLoS ONE*, *G3*, *Evolution*, *IEEE/ACM Transactions on Computational Biology and Bioinformatics*, *EMPH: Evolution, Medicine & Public Health*, *Ecological Modelling*, *Human Population Genetics and Genomics* |
| | 2016 | Committee Member & Presenter, Stanford Postdoc Pedagogy Journal Club |
| | 2015-2016 | Committee Member, Stanford CEHG diversity outreach committee |

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| **Mentorship** | 2024- | Undergrad research mentor for Stephen Burchfield, Tufts Biology Dept |
| | 2024- | Postdoc mentor for Dr. Daniel Wuitchik, Tufts Biology Dept |
| | 2023- | Graduate co-mentor for Kaitlyn Coburn, Tufts Biology Dept |
| | 2022-2024 | Undergrad research mentor for Alex Hovsepian, Tufts Biology Dept & Summer Fellow |
| | 2022-2024 | Undergrad research mentor for Daphne Garcia, Tufts Biology Dept & Summer Fellow |
| | 2022 | Undergrad research mentor for Alison Leav, Tufts Biology Dept |
| | 2021-2022 | Postdoc mentor for Dr. Adam Pepi, Tufts Biology Dept |
| | 2021- | PhD mentor for Kasturi Lele, Tufts Biology Dept |
| | 2021- | PhD mentor for Alejandro Calderon, Tufts Biology Dept |
| | 2021-2023 | PhD mentor for SJ McGeady, Tufts Biology Dept (graduated with MS) |
| | 2021 | Summer research mentor for Alice Lau, Tufts MS student and TIE Environmental Fellow |
| | 2021 | Summer research mentor for SJ McGeady, incoming Tufts PhD student |
| | 2016 | Summer research co-mentor for Alan Aw, Stanford undergraduate |
| | 2013 | Summer research co-mentor for Isela Hernandez, Biological Health Sciences Internship Program |
| | 2010 | Summer research co-mentor for Sam Neal, Summer Link High School Program |

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| **Competitive Funding** | 2022 | Tufts Springboard Award, *Disciplinary Identity Development in Computational Biology* ($15,000) |
| | 2016-2018 | NIH IRACDA Fellow, Stanford & SJSU ($53,600 per year) |
| | 2016-2017 | Stanford Teaching & Mentoring Academy Award ($6,870) |
| | 2015-2016 | Stanford CEHG Postdoctoral Fellowship ($50,000) |
| | 2012-2014 | UCSF Achievement Rewards for College Scientists Fellow ($12,000 per year) |
| | 2014 | UCSF Discovery Fellow ($4,000 for travel & research over 2 years) |

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| **News coverage** | 2026 | Lawrence interviewed on People Behind the Science |
| | 2026 | *Tufts Now* on microbial community assembly |
| | 2019 | *Stanford News* on priority effects in grasslands |

**Selected presentations**

[1] Integrating genomic data and evolutionary models in ecological predictions. *Tufts University, Medford, MA*. **Seminar**, 2026.

[2] Interrogating the Role of Rare Alleles in Resilience to Environmental Change. *Organismal Resilience Meeting of the Linnean Society, London, England.* **Talk** (2025).

[3] The role of mutation spectra in determining ecological outcomes. *Departmental Seminar, UMass Dartmouth, Darmouth,*

*MA*. **Seminar** (2025).

[4] Multimodal selection and the evolution of thermal performance on a warming planet. *American Society of Naturalists Meeting, Monterey, CA*. **Talk** (2025).

[5] Integrating ecological processes into evolutionary models & inference. *UNCC Bioinformatics Departmenal Seminar, Charlotte, NC*. **Seminar** (2024).

[6] The influential role of physics-based methods in evolutionary biology. *Carleton College, Northfield, MN*. **Seminar**, 2024.

[7] What can we learn by integrating ecological processes into evolutionary inference?. *Cornell University, Ithaca, NY*. **Seminar**, 2024.

[8] Eco-evolutionary perspectives on microbial interactions, community assembly, and adaptation. *UMass Chan Medical School, Department of Microbiology and Physiological Systems, Worcester, MA*. **Seminar**, 2023.

[9] Youniss Family Assistant Professorship in Innovation: Evolutionary genetics as a prospective discipline. *Tufts University, Medford, MA*. **Seminar**, 2023.

[10] Evolutionary genetics of costly adaptation to pathogens. *American Naturalist, Asilomar, CA*. **Talk**, 2023.

[11] Infectious disease and costly resistance in evolutionary genetics. *Tufts Biology Departmental Retreat, Medford, MA*. **Talk** (2022).

[12] Back to the future: the population genetics of adaptation in our rapidly changing world. *CSU East Bay, Hayward, CA*. **Seminar**, 2020.

[13] Adaptation and constraint through the lens of genome-wide association studies. *University of Wisconsin, Madison, WI*. **Seminar**, 2020.

[14] Adaptation and constraint through the lens of genome-wide association studies. *Temple University, Philadelphia, PA*. **Seminar**, 2020.

[15] Adaptation and constraint through the lens of genome-wide association studies. *Tufts University, Somerville, MA*. **Seminar**, 2020.

[16] Adaptation and constraint through the lens of genome-wide association studies. *Vanderbilt University, Nashville, TN*. **Seminar**, 2020.

[17] Evolutionary perspectives on human complex traits and missing heritability. *CSUN, Northridge, CA*. **Seminar**, 2019.

[18] Can modern genomic sequence data fulfill their evolutionary potential?. *Indiana University, Bloomington, IN*. **Seminar**, 2019.

[19] Statistical inference of evolutionary processes from genomic data. *Macalester College, St. Paul, MN*. **Seminar**, 2019.

[20] Can modern genomic sequence data fulfill their evolutionary potential?. *Cornell University, Ithaca, NY*. **Seminar**, 2019.

[21] Exploiting selection at linked sites to infer the rate and strength of adaptation. *Evolution Meeting, Providence, RI*. **Talk**, 2019.

[22] Genome-scale inference of adaptive evolution: new approaches for answering old questions. *Boise State University, Boise, ID*. **Seminar**, 2019.

[23] Genome-scale inference of adaptive evolution: new approaches for answering old questions. *Chapman University, Orange, CA*. **Seminar**, 2019.

[24] Evolutionary processes shaping the human genome. *Linfield College, McMinnville, OR*. **Seminar**, 2018.

[25] Scientific teaching workshops for Stanford postdocs. *Stanford Education Day, Stanford, CA*. **Talk**, 2018.

[26] Modulation of adaptation rate by background selection in the human genome. *Bay Area Population Genomics Meeting,*

*Santa Cruz, CA*. **Talk**, 2018.

[27] Modulation of adaptation rate by background selection in the human genome. *American Society of Naturalists Meeting, Monterey, CA*. **Talk** (2018).

[28] Designing and implementing scientific teaching workshops for postdocs. *Stanford Teaching and Mentoring Academy seminar series, Stanford, CA*. **Talk** (2018).

[29] An analytical upper bound on the number of loci required for all splits of a species tree to appear in a set of gene trees. *RECOMB Comparative Genomics Meeting, Montreal, Canada*. **Talk** (2016).

[30] Explosive growth and the genetic architecture of polygenic traits under selection. *American Society of Naturalists Meeting, Monterey, CA*. **Talk** (2016).

[31] Selection and explosive growth may hamper the performance of rare variant association tests. *Bay Area Population Genomics Meeting, Stanford, CA*. **Talk** (2015).

[32] Recent demography and natural selection hamper the power of rare variant association tests. *American Society of Human Genetics Meeting, San Diego, CA*. **Talk** (2014).

[33] Model-based simulations of selection and demography with implications for heritable phenotypes and rare variant association tests. *UC Berkeley Center for Theoretical Evolutionary Genomics, Berkeley, CA*. **Talk** (2014).

[34] Simulations and inference of simultaneous positive and negative selection. *Society of Molecular Biology and Evolution Meeting, San Juan, Puerto Rico*. **Poster** (2014).

[35] Parameter rescaling for forward simulations of recurrent hitchhiking. *Bay Area Population Genomics Meeting, San Francisco, CA*. **Poster** (2013).

[36] Forward simulations of recurrent selection and demographics with rescaled parameters. *American Society of Human Genetics Meeting, Boston, MA*. **Poster** (2013).

[37] Accurate pedigree-based imputation. *Department of Human Genetics Seminar, University of Chicago, Chicago, IL*. **Talk** (2011).

[38] Simultaneous measurement of myosin-II kinetics and cellular traction forces. *HHMI Interfaces Scholars Meeting, Chevy Chase, MD*. **Poster** (2008).